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M 202 CONDENSER

1. SCOPE OF WORK

The Bidder shall supply the following in accordance with **Part 4, Subsection 1.2 "Summary of Work"**.

- (1) One (1) Surface Condenser (Double Pass, divided water box, located below steam turbine).
- (2) Debris Filters
- (3) One (1) Condenser Tube Cleaning System.
- (4) Electrical Equipment and Instrumentation

2. CODES AND STANDARDS

The Bidder shall comply with the requirements of **Part 4, Subsection 2.5 "Codes and Standards"**. The codes and standards of the following organizations extracted from **Part 4, Subsection 2.5 "Codes and Standards"** or alternatively international equivalent standards as specifically applicable to the manufacturer's design and testing of the equipment included in this Specification.

ASME	American Society of Mechanical Engineers.
ANSI	American National Standards Institute.
ASTM	American Society of Testing and Materials.
BS	British Standard
DIN	German Institute for Standardization
EN	European Committee for Standardization
HEI	Heat Exchanger Institute.
IEC	International Electrotechnical Commission.
JIS	Japan Industrial Standard

NEMA	National Electrical Manufacturers Association.
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3. DESIGN AND OPERATING CONDITIONS

3.1 Design Operating Data

Source of circulating water	Sea Water
Cleanliness factor	85 %
Circulating water temperature inlet	32.5 °C.
Maximum dissolved oxygen permitted in condensate	0.015 mg/kg
Maximum tube velocity	2.4 m/s.
Design pressure	The condenser shall be designed for the maximum allowable back pressure at which the turbine can be operated continuously without any restrictions.
Tube size	25 mm outside diameter
Tube wall thickness	0.5 mm minimum thickness.

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Minimum shell and hotwell design pressure: 1 bar(g) or the maximum pressure due to hydrostatic test (for the hydrostatic test condenser will be filled with water up to the turbine exhaust outlet), whichever is greater and full vacuum conditions.

Water box design pressure shall be the pressure at the bottom of the water box corresponding to circulating water pump shut off head, rounded off to the next higher 0.5 bar plus an allowance for water hammer surge as verified by Bidder's calculations.

Condenser shall be designed to receive and condense the whole of the steam from the steam turbine, including the by-pass system, under all conditions of load and operation.

The cooling water flow through the condenser of each block, with using 2 cooling water pumps are in operation, shall be adequate for the following conditions:

- (1) The Steam Turbine Generator is operating at peak load with Gas Turbine at 100% load.
- (2) Maximum condenser pressure allowed for continuous operation (Bidder to advise).
- (3) Maximum cooling water temperature rise through the condenser shall not exceed 7 °C with Gas Turbine at 100% load.

The maximum cooling water outlet temperature at the discharge condenser shall not exceed 40°C with condition all cooling water pumps are in operation.

The surface area of the condenser shall be optimized by the Bidder considering the parameters given above, the hydraulic gradient of the CW system, the economical aspects, etc. This study shall be submitted to the Owner for approval.

C.W. inlet line to Condenser - Pipe diameter - Bidder to determine based on a maximum pipe velocity of 2.2 m/s.

3.2 Sea Water Analysis

Refer to **Part 4, Subsection 1.1.**

3.3 Circulating Water (CW) Pump Trips

The condenser shall be capable of sustained operation at part load with one CW pumps should the second operating pump trip.

4. DETAILED REQUIREMENTS

4.1 Design and Constructions

The condenser shall be designed and constructed in accordance with HEI Standards for steam surface condensers **and shall be located below steam**

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turbine and surface chosen with respect to the appropriate cleanliness factors/fouling resistance.

Condenser shall be of the horizontal surface type, with internal air cooling and removal section(s) and auxiliaries as hereinafter specified.

All parts of condenser shall be designed to withstand the most severe conditions to which they will be subjected, with a factor of safety commensurate with the function, service, and/or material of each part. Structural design shall be in accordance with sound commercial engineering practice for the service required.

The condenser shall be located beneath the turbine exhaust, and shall be designed for a minimum width and for installation within a reinforced concrete turbine foundation.

The surface condenser shall be designed to accommodate the turbine bypass system without thermal stress, drain attack, vacuum trip, etc.

The condenser shall be completely shop tubed and assembled.

Shell, water boxes and heads shall be properly ribbed, braced and otherwise reinforced as required.

Materials: Bidder shall be responsible for providing pumps of design and materials best suited for the specified service conditions. If in the opinion of the Bidder the materials, where specified, are not adequate, Bidder shall, in his proposal, recommend those materials which he deems most suitable for the service.

(a) Water boxes and water box covers	Carbon Steel ASTM A285 Gr C, Flange Quality, copper bearing. Internally protected with 5 mm thick rubber lining.
(b) Shell plate (including neck and hotwell)	Carbon Steel ASTM A285 Gr C, Flange Quality
(c) Tube sheets	Titanium clad
(d) Tube support plates	Carbon Steel ASTM A285 Gr C, Flange Quality
(e) Tubes	Titanium Grade 2

(f) Pipe used for internal or external supporting and reinforcing members	ASTM A53 Grade B.
(g) External bolts and nuts	Alloy steel ASTM A194.
(h) Bolts and nuts exposed to sea water or steam	Non corrosive material or with suitable surface protection for the service.
(i) Gaskets	Materials most suitable for the service.
(j) Flexible Joints	Neoprene

The condenser shall be of all welded construction in so far as is practicable.

Joints shall be full penetration fusion welded using an arc welding process conforming to the requirements of the ASME Boiler and Pressure Vessel Code, Section VIII, Division I. Double-welded butt joints shall be used wherever possible.

Adequate provisions shall be made for expansion and contraction. All parts shall be readily accessible for inspection and repair.

Manhole covers and access opening covers on condenser shell, water boxes, hotwell etc. shall be equipped with quick opening devices and means for locking in the open position

4.2 Shell and Hotwell

Joints between shell, tube sheets and water boxes shall be flanged with external bolting and made up with heavy suitable gaskets to assure tight steam and water seals. Alternatively the joints can be welded.

At least one manhole with gasketed cover shall be provided for access to the steam space between the steam inlet and tubes. The manhole size shall be large enough to permit the passage of materials required for the replacement of the expansion joint between the, turbine exhaust and condenser (if applicable). At least one hinged manhole with gasketed cover shall be provided for access to the hotwell. The size, number and location of manholes shall be shown on Bidder's drawings.

All inlets on condenser shell for turbine bypass that may be located opposite the tube bank shall be provided with baffles or perforated distribution headers to prevent erosion of tubes.

Provisions shall be made in the design to absorb differential expansion between tubes and shell without permitting leakage of circulating water into the steam spaces.

The condenser design shall provide for adequate sloping of tubes to assure complete drainage of tubes.

The air removal section shall be designed to prevent any stagnant accumulation of air and non condensable gases. The air removal section shall be arranged so that it extends for the length of the condenser tubes in order to minimize the number of tubes in the air removal section and arranged so that it has an adequate number of transverse baffles. Hoods, baffles and offtake piping in the air removal section shall be 316 stainless steel. For air removal system refer to **Part 4 Section 3, M-203**.

The condenser hotwell shall have ample capacity to provide proper submergence and volume for smooth operation of the condensate pumps when the unit is operating at maximum full load. In no case shall the hotwell be designed for less than two minutes capacity under conditions of maximum full load.

Provisions shall be made to prevent rust and foreign matter from passing out of the condenser unit through the hotwell discharge outlets. In general, this shall consist of a dam or dirt collar at each outlet to prevent foreign matter that has settled on the hotwell bottom from being carried into the outlets.

The condenser design shall provide for the entrance loss of condensate into the pump suction piping (with minimum hotwell level established allowing for condensate pump suction piping entrance loss). Anti-vortex vanes shall be provided in the hotwell outlets, if required. The hotwell levels and condensate pump suction piping arrangement shall be compatible with the condensate pump hydraulic requirements.

The location and design of steam and water dumps to the condenser shell shall be in accordance with ASME Standard TDP-1, Part 1. Alternatively a vacuum

flash tank can be used for steam turbine internal drains and an atmospheric drain tank for external drains.

Means for the detection of sea water leakage into the steam space at the ends of the tube sheets shall be provided for connection via sampling pumps to sampling panels.

4.3 Water Boxes

The water boxes shall be of good hydraulic design and liberal in size to provide low water velocities and optimum entrance conditions to the tubes.

Water boxes shall be welded, steel plate construction. Water box stiffeners shall be located on the outside.

Water boxes shall be provided with manholes and access openings as follows:

- (1) Manholes with quick opening type hinged covers and lugs for locking in the open position.
- (2) Access openings, with bolted covers, to permit replacement of tubes in air removal sections.

The size, number and location of manholes and access openings shall be shown on Bidder's drawings.

Water boxes shall be provided with suitable vent and drain connections.

Each section of water box shall be provided with eye bolts or lugs for lifting. Jacking screws shall be provided for breaking the joints between heads and water boxes.

The design shall ensure that there is no danger of system becoming air bound, resulting in the lost of the designed siphon head. Should air removal equipment be required to maintain the siphon head, the Bidder shall provide 2x100% air removal equipment as part of his system. Such equipment shall be complete with start/stop level switches installed in vacuum level control tanks complete with fittings and valves. For air removal system refer to **Part 4 Section 3, M-203**.

Circulating water openings at the water box bottom shall be covered with wide spaced removable grating for personnel protection.

4.4 Tube Sheets

The tube sheets shall be of thickness conforming to the heat Exchange Institute Standard and shall have reamed and countersunk holes for the tubes.

Tube vibration shall be minimized by suitable support of the pipes.

4.5 Condenser Supports

The condenser shall be supported by the building foundation.

All required thrust blocks, slide plates and sole plates guide brackets, shall be furnished by Bidder.

Bidder shall provide lubricated metal slide plates, guides and keeper bars, at all condenser supports as far as required.

Any anchor bolts required for condenser supporting foundations will be furnished by Bidder.

The Bidder shall provide full details of the method he proposes to accommodate thermal expansion which shall be to the approval of the Owner.

4.6 Turbine-Condenser Joint

The expansion joint between the turbine exhaust outlet and the condenser neck inlet, shall be welded.

4.7 Tubes

Tubes shall be suitable for rolling into tube sheets without causing any injury or harmful stresses in the tubes. Tubes shall be expanded into the tube sheets and seal welded at both ends. The tube ends shall be profiled to suit the on line cleaning system offered.

The exposed tubes on the top rows of each nest and at the steam lanes shall be adequately protected from impingement damage by the turbine exhaust steam.

4.8 Inlet and Outlet

Connections shall be provided on the water and steam sides as required.

To prevent tube erosion, inlets on condenser shell for the admission of drains and vents from equipment and piping systems shall be provided with alloy steel plate baffles or alloy steel perforated distribution headers.

Nozzle locations shall be shown on Bidder's drawings.

Connections shall be as follows, except where otherwise approved by the Owner:

(1) Circulating water inlets and outlets:

Steel water boxes - Steel flanged, suitable for Class 125 connections.

(2) Flanged connections where specifically required on steel parts (other than circulating water connections):

Steel nozzles with Class 150 welding neck or slip-on flanges.

4.9 Debris Filter

The debris filters shall be provided at the circulating water connection of each inlet water box and if required at the connection to closed cooling water heat exchangers

The condenser circulating water inlet, outlet and debris discharge valves shall be motor operated.

The debris filter shall successfully remove debris from the circulating water inlet piping under normal filter operating conditions.

No debris larger than the filter basket perforations, whose hole size is preferably 5 mm, shall pass through the debris filters.

The Bidder shall certify that the manufacturer of debris filters has carried out at least 5 installations which are successfully in operation for minimum 5 years. These reference filters must have a size of minimum 50”.

The debris filter system shall be designed to operate automatically, successfully removing all types of debris from condenser circulating water inlet piping.

Debris shall be removed from the circulating water while the generating unit is on-line, and will require no load reduction, and shall do so without operator's attention.

During flushing not more than 4% of the main cooling water flow may be discharged.

The flushing cycle shall be optimized by a filter optimizer.

The filter element shall have satisfactory filtration and prevent any debris to pass by. The filter element shall be of perforated type.

For the purpose of easy maintenance, the filter element shall be fixed by bolts and not by mean of welding.

The filter element shall be designed to withstand a pressure loss of minimum 0.7 bar without restrictions of plant operation.

The debris filter shall be designed with maximum friction coefficient of 2.4 at clean condition.

4.10 Condensate Pump Suction Piping

Strainer screens shall be of the removable single basket type with quick removal cover; made from heavy gauge perforated Type 304 stainless steel sheet with 3.2 mm perforations. The open area of the screen shall, be equal to not less than 2.5 times the cross sectional area of the corresponding pipe size

Welded joints shall be used in this piping, except at connection to equipment, at flanged valves, expansion joints and at strainers, where flanged joints must necessarily be used

Expansion joints shall be rubber designed for full vacuum. The exact type of joint used, and the details of joint reinforcement etc., shall be subject to Owner approval.

4.11 Shop Assembly

The condenser shall be shop-assembled to the maximum extent dictated by shop manufacturing limitations or design and transport considerations.

4.12 Condenser Tube Cleaning System

The Bidder shall be capable of providing single source responsibility for the elimination of both micro and macro condenser fouling, the Bidder shall be capable of furnishing ball-type condenser tube cleaning system, in addition to debris filters.

The Condenser Tube Cleaning System shall be an intermittent on-line system using sponge rubber cleaning balls. The condenser shall be provided with a complete ball cleaning system servicing all water boxes. The condenser tube cleaning system shall work continuously cleaning intermittently each half of the condenser for a period that can be established by the operators. In the half of the condenser that is being cleaned at least one cleaning ball shall pass through each tube on an average of once every 5 minutes. The ball recirculating system shall operate on a closed cycle around the condenser. The cleaning balls shall be injected at the inlet of the condenser and after passing through the tubes shall be collected at the outlet by special strainer. A ball recirculating pump shall be provided to recirculate the cleaning balls. All necessary pipes, valves, and controls for this system shall be provided.

4.13 Cleaning, Coating and Painting

(1) Cleaning

Interior and exterior surfaces of the condenser and appurtenances, including shell, tube sheets, hotwell, tube support plates, baffles, water boxes, etc., shall be thoroughly cleaned as required by metallic shot or grit blasting so as to leave clean metal surfaces. Inside surfaces of condenser steam spaces shall not be painted or coated.

(2) Lining

The condenser water box interior shall be rubber lined. The material and method of application shall be to the approval of the Owner.

(3) Painting

All exterior steel plate and rolled steel surfaces shall be painted in accordance with **Part 4 Section 3, S-420 and S-430**

Pipe flanges, fittings, and valves shall be given a coat of heat-resistant aluminum paint, or other paint suitable for the service.

5. TESTING

5.1 Performance Test

After equipment is put in service the Bidder shall run a check test to ascertain whether the equipment is performing properly in every respect. Testing shall be in accordance with the requirements of Part 4, Subsection 2.6 Performance Testing and Guarantees .

Any deficiency brought out by this test shall be promptly corrected by the Bidder at no cost to the Owner.

5.2 Shop Test

All water boxes shall be hydrostatically tested to 1.5 times their maximum allowable working pressure, for a minimum duration of 60 minutes. Inspection and testing should be in line with **Part 4, Section 3, M-014**.

5.3 Field Tightness Tests

After erection the equipment shall be tested for tightness as follows

- (1) The steam space (s) of the condenser shell (s) shall be tested for leakage using hydrostatic-fluorescent inspection (black light). The steam space (s) shall be completely filled with a concentration of water and water soluble fluorescence up to the steam inlet connection and proven tight. The water boxes shall be empty during this test.
- (2) The water boxes shall be hydrostatically tested to a pressure which is 1.5 times the water box design pressure, but in no case less than 1.7 bar(g) measured at the bottom of the water box. The steam space shall be empty during this test. The test shall have a minimum duration of 60 minutes.

6. QUALITY ASSURANCE/QUALITY CONTROL (QA/QC)

Quality control shall conform to the requirements of **Part 4, Subsection 2.7**

7. INSTRUMENTATION AND CONTROLS

7.1 Instrumentation

The surface condenser shall be provided with the following instruments

- (1) armored level gauge glass
- (2) level switch with alarm
- (3) level transmitter with remote indicator
- (4) pressure gauge (inc manometer)
- (5) pressure transmitter with remote indicator
- (6) thermometer
- (7) level controller and control valves
- (8) flow meter for make up water
- (9) low-low level switch

7.2 Design Requirements

The Bidder shall be responsible for the design, supply and installation of complete instrumentation and control system and devices for the condenser tube cleaning system including but not limited to those listed and described in this specification.

The local control panel shall contain all control and monitoring devices which do not need to be installed at a local measuring location, and include all operator interfaces such as indicators, push buttons, lights, control switches and alarms.

Refer to **Part 4, Section 3, I-002 "Local Control Panels"** for the design requirements for the local control panels, and **Part 4, Section 3, I-001 "Instrumentation and Control"** for the design requirements for Instrumentation and Control.

8. ELECTRICAL SPECIFICATIONS

8.1 General

Electrical equipment and accessories

The Bidder shall furnish and install all electrical equipment and accessories required for the condenser and its tube cleaning system.

For general electrical requirements, see **Part 4, Section 3, E-001**

Motors

For detailed electrical requirements, see **Part 4, Section 3, E-210** for Low Voltage Motors.

8.2 Power Panel

For detailed electrical requirements, see **Part 4, Section 3, E-001 and E-280**.

8.3 Conduits, Cables and Wiring

Detailed electrical requirement, see **Part 4, Section 3, E-232 and E-230**.

8.4 Cathodic Protection System

For detailed electrical requirements, see **Part 4, Section 3, E-260**.

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